

Лабораторная работа №4

Работа с графикой в LaTeX

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06 декабря 2025

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Цель работы

- Изучить базовую структуру графических компонентов LaTeX
- Освоить подключение изображений через **graphicx**
- Научиться работать с плавающими объектами (**figure**)
- Освоить масштабирование, вращение, управление положением изображений
- Освоить создание собственных float-окружений

Ход выполнения

- Создан каталог для упражнений главы 4
- Проверена структура директории
- Загружены тестовые изображения

```
PS C:\WINDOWS\system32> cd $HOME\Documents\latex-exercises
PS C:\Users\ousma\Documents\latex-exercises> mkdir chapitre-4-graphics

Répertoire : C:\Users\ousma\Documents\latex-exercises

Mode                LastWriteTime         Length Name
----                -
d-----          06/10/2025   00:41             chapitre-4-graphics

PS C:\Users\ousma\Documents\latex-exercises> cd chapitre-4-graphics
PS C:\Users\ousma\Documents\latex-exercises\chapitre-4-graphics> |
```

Рис. 1: Создание каталога

- В каталоге обнаружен файл `simple-image.pdf`
- Подготовлен файл для упражнения 4.1

```
Répertoire : C:\Users\ousma\Documents\latex-exercises\chapter-4-graphics

Mode                LastWriteTime         Length Name
----                -
-a-----          23/10/2025    02:39             9050 simple-image.pdf

PS C:\Users\ousma\Documents\latex-exercises\chapter-4-graphics> @'
>> \documentclass{article}
>> \usepackage[T1]{fontenc}
>> \usepackage{graphicx}
>>
```

Рис. 2: Проверка содержимого каталога

Упражнение 4.1

- Подключены пакеты `graphicx`
- Добавлены примеры изменения размеров изображений
- Документ успешно скомпилирован через `pdflatex`

```
>> \end{center}
>>
>> \end{document}
>> '@ | Out-File -FilePath "exercise-4-1.tex" -Encoding UTF8
PS C:\Users\ousma\Documents\latex-exercises\chapter-4-graphics> pdflatex exercise-4-1.tex
This is pdfTeX, Version 3.141592653-2.6-1.40.28 (TeX Live 2025) (preloaded format=pdflatex)
 restricted \write18 enabled.
entering extended mode
(./exercise-4-1.tex
LaTeX2e <2025-06-01> patch level 1
L3 programming layer <2025-09-02>
(c:/texlive/2025/texmf-dist/tex/latex/base/article.cls
Document Class: article 2025/01/22 v1.4n Standard LaTeX document class
(c:/texlive/2025/texmf-dist/tex/latex/base/size10.clo)
(c:/texlive/2025/texmf-dist/tex/latex/base/fonts-std)
```

Рис. 3: Создание exercise-4-1.tex

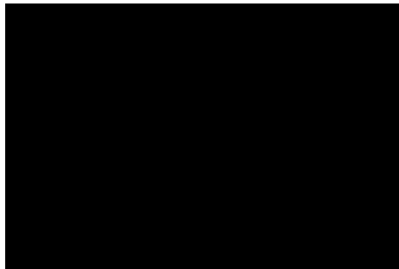
Exercise 4.1 - Altering Graphic Appearance

Part 1: Basic Sizing

1.1 Image at 25% of text width:



1.2 Image at 50% of text height:



1.3 Image with fixed width of 4cm:



Part 2: Rotation and Scaling

2.1 Image rotated 30 degrees:



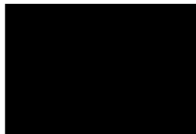
2.2 Image rotated -45 degrees:



2.3 Image scaled to 0.3:



2.4 Image scaled to 1.5:



Part 3: Parameter Combinations

3.1 Rotation + resizing:

3.2 Testing aspect ratio:



Проверка сгенерированных файлов

Автоматически созданные файлы

- .aux
- .log
- ИТОВОВЫЙ .pdf
- ИСХОДНЫЙ .tex

```
PS C:\Users\ousma\Documents\latex-exercises\chapter-4-graphics> start exercise-4-1.pdf
PS C:\Users\ousma\Documents\latex-exercises\chapter-4-graphics> dir
```

Répertoire : C:\Users\ousma\Documents\latex-exercises\chapter-4-graphics

Mode	LastWriteTime		Length	Name
----	-----	-----	-----	----
-a----	23/10/2025	02:41	32	exercise-4-1.aux
-a----	23/10/2025	02:41	6898	exercise-4-1.log
-a----	23/10/2025	02:41	34763	exercise-4-1.pdf
-a----	23/10/2025	02:40	1382	exercise-4-1.tex
-a----	23/10/2025	02:39	32	simple-image.aux
-a----	23/10/2025	02:39	4499	simple-image.log
-a----	23/10/2025	02:39	9050	simple-image.pdf
-a----	23/10/2025	02:39	141	simple-image.tex
-a----	23/10/2025	02:33	8	test-images.aux
-a----	23/10/2025	02:33	4592	test-images.log
-a----	23/10/2025	02:28	520	test-images.tex

Упражнение 4.2 — работа с figure

- Использованы параметры: `h`, `t`, `b`
- Проверено автоматическое размещение LaTeX

Exercise 4.2 - Making Images Float

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

Floating image with automatic placement

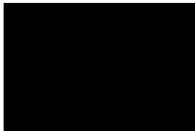


Figure 1: A floating image with automatic placement

Nulla malesuada porttitor diam. Donec felis erat, congue non, volutpat at,

pellentesque felis eu massa.

Quisque ullamcorper placerat ipsum. Cras nibh. Morbi vel justo vitae lacus tincidunt ultrices. Lorem ipsum dolor sit amet, consectetur adipiscing elit. In hac habitasse platea dictumst. Integer tempus convallis augue. Etiam facilisis. Nunc elementum fermentum wisi. Aenean placerat. Ut imperdiet, enim sed gravida sollicitudin, felis odio placerat quam, ac pulvinar elit purus eget enim. Nunc vitae tortor. Proin tempus nibh sit amet nisl. Vivamus quis tortor vitae risus porta vehicula.

Different placement options



Figure 2: Option [h] - Here if possible

Fusce mauris. Vestibulum luctus nibh at lectus. Sed bibendum, nulla a faucibus semper, leo velit ultricies tellus, ac venenatis arcu wisi vel nisl. Vestibulum diam. Aliquam pellentesque, augue quis sagittis posuere, turpis lacus congue quam, in hendrerit risus eros eget felis. Maecenas eget erat in sapien mattis porttitor. Vestibulum porttitor. Nulla facilisi. Sed a turpis eu lacus commodo facilisis. Morbi fringilla, wisi in dignissim interdum, justo lectus sagittis dui, et vehicula libero dui cursus dui. Mauris tempor ligula sed lacus. Duis cursus enim ut augue. Cras ac magna. Cras nulla. Nulla egestas. Curabitur a leo. Quisque egestas wisi eget nunc. Nam feugiat lacus vel est. Curabitur consectetur.

Suspendisse vel felis. Ut lorem lorem, interdum eu, tincidunt sit amet, laoreet vitae, arcu. Aenean faucibus pede eu ante. Praesent enim elit, rutrum at, molestie non, nonummy vel, nisl. Ut lectus eros, malesuada sit amet, fermentum eu, sodales cursus, magna. Donec eu purus. Quisque vehicula, urna sed ultricies auctor, pede lorem egestas dui, et convallis elit erat sed nulla. Donec luctus. Curabitur et nunc. Aliquam dolor odio, commodo pretium, ultricies non, pharetra in, velit. Integer arcu est, nonummy in, fermentum faucibus, egestas vel, odio.

Sed commodo posuere pede. Mauris ut est. Ut quis purus. Sed ac odio. Sed vehicula hendrerit sem. Duis non odio. Morbi ut dui. Sed accumsan risus eget odio. In hac habitasse platea dictumst. Pellentesque non elit. Fusce sed justo eu urna porta tincidunt. Mauris felis odio, sollicitudin sed, volutpat a, ornare ac, erat. Morbi quis dolor. Donec pellentesque, erat ac sagittis semper, nunc dui lobortis purus, quis congue purus metus ultricies tellus. Proin et quam. Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos



Figure 3: Option [t] - Top of page

hymenaeos. Praesent sapien turpis, fermentum vel, eleifend faucibus, vehicula eu, lacus.

Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Donec odio elit, dictum in, hendrerit sit amet, egestas sed, leo. Praesent feugiat sapien aliquet odio. Integer vitae justo. Aliquam vestibulum fringilla lorem. Sed neque lectus, consectetur at, consectetur sed, eleifend ac, lectus. Nulla facilisi. Pellentesque eget lectus. Proin eu metus. Sed porttitor. In hac habitasse platea dictumst. Suspendisse eu lectus. Ut mi mi, lacinia sit amet, placerat et, mollis vitae, dui. Sed ante tellus, tristique ut, iaculis eu, malesuada ac, dui. Mauris nibh leo, facilisis non, adipiscing quis, ultrices a, dui.



Figure 4: Option [b] - Bottom of page

Установка пакета lipsum

- Используется генерация текстовых вставок
- Пакет успешно установлен

```
PS C:\Users\ousma\Documents\latex-exercises\chapter-4-graphics> tlmgr install lipsum
tlmgr.pl: package repository https://mirror.truenetwork.ru/CTAN/systems/texlive/tlnet (not verified: gpg unavailable)
[1/1, ??:??/??:??] install: lipsum [879k]
running mktexlsr ...
done running mktexlsr.
tlmgr.pl: package log updated: C:/texlive/2025/texmf-var/web2c/tlmgr.log
tlmgr.pl: command log updated: C:/texlive/2025/texmf-var/web2c/tlmgr-commands.log
PS C:\Users\ousma\Documents\latex-exercises\chapter-4-graphics> @'
>> \documentclass{article}
...
538366
...

```

Рис. 11: Установка lipsum

Упражнение 4.1 — альтернативный вариант

Exercise 4.1 - Showing Graphic Appearance
Using nested image on the document
#2018-000000



Diagram description

1



Diagram description

2

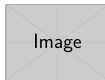


Diagram description

3

```
PS C:\Users\ousma\Documents\latex-exercises\chapter-4-graphics> pdflatex exercise-4-2.tex
>> start exercise-4-2.pdf
This is pdfTeX, Version 3.141592653-2.6-1.40.28 (TeX Live 2025) (preloaded format=pdflatex)
 restricted \write18 enabled.
entering extended mode
(./exercise-4-2.tex
LaTeX2e <2025-06-01> patch level 1
L3 programming layer <2025-09-02>
(c:/texlive/2025/texmf-dist/tex/latex/base/article.cls
Document Class: article 2025/01/22 v1.4n Standard LaTeX document class
(c:/texlive/2025/texmf-dist/tex/latex/base/size10.clo))
(c:/texlive/2025/texmf-dist/tex/latex/base/fontenc.sty)
(c:/texlive/2025/texmf-dist/tex/latex/graphics/graphicx.sty
```

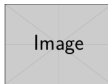
Рис. 13: Компиляция

- Проверить влияние форматов и имён файлов на работу LaTeX

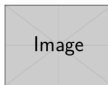
Exercise 4.3 - Naming Graphics Files

Testing file naming

Without extension (LaTeX guesses):



With explicit extension:



Best practices:

- Use simple names: image1.pdf, diagram.png
- Avoid spaces: my_image.jpg not "my image.jpg"
- No special characters or accents
- Descriptive but short names

Recommended file structure:

```
project/  
  main.tex  
  images/  
    diagrams/  
      fig1.pdf  
      fig2.pdf  
    photos/
```

Упражнение 4.6

```
>> pdflatex exercise-4-6.tex
>> start exercise-4-6.pdf
This is pdfTeX, Version 3.141592653-2.6-1.40.28 (TeX Live 2025) (preloaded format=pdflatex)
 restricted \write18 enabled.
entering extended mode
(./exercise-4-6.tex
LaTeX2e <2025-06-01> patch level 1
L3 programming layer <2025-09-02>
(c:/texlive/2025/texmf-dist/tex/latex/base/article.cls
Document Class: article 2025/01/22 v1.4n Standard LaTeX document class
(c:/texlive/2025/texmf-dist/tex/latex/base/size10.clo))
(c:/texlive/2025/texmf-dist/tex/latex/base/fontenc.sty)
(c:/texlive/2025/texmf-dist/tex/latex/graphics/graphicx.sty
(c:/texlive/2025/texmf-dist/tex/latex/graphics/keyval.sty)
(c:/texlive/2025/texmf-dist/tex/latex/graphics/graphics.sty
(c:/texlive/2025/texmf-dist/tex/latex/graphics/trig.sty)
(c:/texlive/2025/texmf-dist/tex/latex/graphics-cfg/graphics.cfg)
(c:/texlive/2025/texmf-dist/tex/latex/graphics-def/pdftex.def)))
(c:/texlive/2025/texmf-dist/tex/latex/lipsum/lipsum.sty
(c:/texlive/2025/texmf-dist/tex/latex/l3packages/l3keys2e/l3keys2e.sty
(c:/texlive/2025/texmf-dist/tex/latex/l3kernel/expl3.sty
(c:/texlive/2025/texmf-dist/tex/latex/l3backend/l3backend-pdftex.def)))
(c:/texlive/2025/texmf-dist/tex/latex/lipsum/lipsum.ltd.tex))
```

Рис. 15: Лог компиляции

Структура каталогов (4.4)

```
PS C:\Users\ousma\Documents\latex-exercises\chapter-4-graphics> Copy-Item "simple-image.pdf" -Destination "images/diagrams/fig1.pdf"
PS C:\Users\ousma\Documents\latex-exercises\chapter-4-graphics> @'
>> \documentclass{article}
>> \usepackage[T1]{fontenc}
>> \usepackage{graphicx}
>>
>> \graphicspath{{images/}{images/diagrams/}{images/photos/}}
>>
>> \begin{document}
>>
>> \section*{Exercise 4.4 - Storing Graphics in Subdirectory}
>>
>> \subsection*{Including from subdirectories}
>>
>> \textbf{From diagrams subdirectory:}
>> \begin{center}
>> \includegraphics[width=0.3\textwidth]{diagrams/fig1}
>> \end{center}
>>
>> \textbf{Using graphicspath (simpler):}
>> \begin{center}
```

Рис. 16: Подготовка каталогов

Exercise 4.4 - Storing Graphics in Subdirectory

Including from subdirectories

From diagrams subdirectory:



Using graphicspath (simpler):



Graphicspath configuration:

```
\graphicspath{
  {images/}
  {images/diagrams/}
  {images/photos/}
}
```

Benefits:

- Clean organization
- Easy maintenance
- Team collaboration
- Backup efficiency

Упражнение 4.5 — создание графики

- PDF, SVG — для векторных
- PNG, JPG — для растровых
- Использование инструментов Inkscape, GIMP и др.

Exercise 4.5 - Producing Graphics

Recommended formats:

- **PDF:** Best for vector graphics, scalable
- **SVG:** Vector format, good for web
- **PNG:** Lossless, good for diagrams
- **JPG:** Lossy, good for photos

Tools for creating graphics:

- **Inkscape:** Vector graphics (SVG, PDF)
- **GIMP:** Raster graphics (PNG, JPG)
- **Matplotlib:** Python plotting (PDF, PNG)
- **TikZ:** Direct LaTeX drawing

Best practices:

- Use PDF for LaTeX when possible
- High resolution for bitmaps (300+ DPI)

Упражнение 4.5 — компиляция

Упражнение 4.5 — компиляция

```
./exercise-4-5.tex
LaTeX2e <2025-06-01> patch level 1
l3 programming layer <2025-09-02>
(c:/texlive/2025/texmf-dist/tex/latex/base/article.cls
Document Class: article 2025/01/22 v1.4n Standard LaTeX document class
(c:/texlive/2025/texmf-dist/tex/latex/base/size10.clo))
(c:/texlive/2025/texmf-dist/tex/latex/base/fontenc.sty)
(c:/texlive/2025/texmf-dist/tex/latex/graphics/graphicx.sty
(c:/texlive/2025/texmf-dist/tex/latex/graphics/keyval.sty)
(c:/texlive/2025/texmf-dist/tex/latex/graphics/graphics.sty
(c:/texlive/2025/texmf-dist/tex/latex/graphics/trig.sty)
(c:/texlive/2025/texmf-dist/tex/latex/graphics-cfg/graphics.cfg)
(c:/texlive/2025/texmf-dist/tex/latex/graphics-def/pdftex.def)))
(c:/texlive/2025/texmf-dist/tex/latex/l3backend/l3backend-pdftex.def)
No file exercise-4-5.aux.
(c:/texlive/2025/texmf-dist/tex/context/base/mkii/supp-pdf.mkii
[Loading MPS to PDF converter (version 2006.09.02).]
) (c:/texlive/2025/texmf-dist/tex/latex/epstopdf-pkg/epstopdf-base.sty
(c:/texlive/2025/texmf-dist/tex/latex/latexconfig/epstopdf-sys.cfg))
[1{c:/texlive/2025/texmf-var/fonts/map/pdftex/updmap/pdftex.map}]
(./exercise-4-5.aux) <c:/Users/ousma/.texlive2025/texmf-var/fonts/pk/ljfour/j
knappen/ec/ecrm1000.600pk> <c:/Users/ousma/.texlive2025/texmf-var/fonts/pk/ljfo
ur/jknappen/ec/ecbx1000.600pk> <c:/Users/ousma/.texlive2025/texmf-var/fonts/pk/
ljfour/jknappen/ec/tcrm1000.600pk> <c:/Users/ousma/.texlive2025/texmf-var/fonts
/pk/ljfour/jknappen/ec/ecbx1200.600pk> <c:/Users/ousma/.texlive2025/texmf-var/f
onts/pk/ljfour/jknappen/ec/ecbx1440.600pk>
Output written on exercise-4-5.pdf (1 page, 34122 bytes).
Transcript written on exercise-4-5.log.
```

Рис. 19: Лог компиляции

Проверка simple-image.tex

```
PS C:\Users\ousma\Documents\latex-exercises\chapter-4-graphics> '@'
>> \documentclass{article}
>> \usepackage[paperwidth=5cm,paperheight=3cm,margin=0pt]{geometry}
>> \begin{document}
>> \rule{5cm}{3cm}
>> \end{document}
>> '@ | Out-File -FilePath "simple-image.tex" -Encoding UTF8
>>
>> pdflatex simple-image.tex
This is pdfTeX, Version 3.141592653-2.6-1.40.28 (TeX Live 2025) (preloaded format=pdflatex)
 restricted \write18 enabled.
entering extended mode
(./simple-image.tex
LaTeX2e <2025-06-01> patch level 1
L3 programming layer <2025-09-02>
(c:/texlive/2025/texmf-dist/tex/latex/base/article.cls
Document Class: article 2025/01/22 v1.4n Standard LaTeX document class
(c:/texlive/2025/texmf-dist/tex/latex/base/size10.clo))
(c:/texlive/2025/texmf-dist/tex/latex/geometry/geometry.sty
(c:/texlive/2025/texmf-dist/tex/latex/graphics/keyval.sty)
(c:/texlive/2025/texmf-dist/tex/generic/iftex/iftex.sty
(c:/texlive/2025/texmf-dist/tex/generic/iftex/iftex.sty)))
(c:/texlive/2025/texmf-dist/tex/latex/l3backend/l3backend-pdf.tex.def)
No file simple-image.aux.
*geometry* driver: auto-detecting
*geometry* detected driver: pdftex

Overfull \hbox (15.0pt too wide) in paragraph at lines 4--5
[1]
```

Упражнение 4.6

```
PS C:\Users\ousma\Documents\latex-exercises\chapter-4-graphics> @'  
>> \documentclass{article}  
>> \usepackage[T1]{fontenc}  
>> \usepackage{graphicx}  
>> \usepackage{lipsum}  
>> \usepackage{float}  
>>  
>> \begin{document}  
>>  
>> \section*{Exercise 4.6 - Placing Floats}  
>>  
>> \subsection*{Automatic float placement}  
>>  
>> \lipsum[1]  
>>  
>> \begin{figure}[htbp]
```

Рис. 21: Создание документа

```
(./exercise-4-6.tex
LaTeX2e <2025-06-01> patch level 1
L3 programming layer <2025-09-02>
(c:/texlive/2025/texmf-dist/tex/latex/base/article.cls
Document Class: article 2025/01/22 v1.4n Standard LaTeX document class
(c:/texlive/2025/texmf-dist/tex/latex/base/size10.clo))
(c:/texlive/2025/texmf-dist/tex/latex/base/fontenc.sty)
(c:/texlive/2025/texmf-dist/tex/latex/graphics/graphicx.sty
(c:/texlive/2025/texmf-dist/tex/latex/graphics/keyval.sty)
(c:/texlive/2025/texmf-dist/tex/latex/graphics/graphics.sty
(c:/texlive/2025/texmf-dist/tex/latex/graphics/trig.sty)
(c:/texlive/2025/texmf-dist/tex/latex/graphics-cfg/graphics.cfg)
(c:/texlive/2025/texmf-dist/tex/latex/graphics-def/pdftex.def)))
(c:/texlive/2025/texmf-dist/tex/latex/lipsum/lipsum.sty
(c:/texlive/2025/texmf-dist/tex/latex/l3packages/l3keys2e/l3keys2e.sty
(c:/texlive/2025/texmf-dist/tex/latex/l3kernel/expl3.sty
(c:/texlive/2025/texmf-dist/tex/latex/l3backend/l3backend-pdftex.def)))
```

Рис. 22: Компиляция

Automatic float placement

Automatic float placement



Figure 1: Automatic placement with [atop]

[illegible]

Figure 2. Forced placement with β_1 (requires float, perhaps)

First placement options

- 26/40

```
PS C:\Users\ousma\Documents\latex-exercises\chapter-4-graphics> @'  
>> \documentclass{article}  
>> \usepackage[T1]{fontenc}  
>> \usepackage{graphicx}  
>> \usepackage{trivfloat}  
>>  
>> \trivfloat{photo}  
>> \trivfloat{diagram}  
>>  
>> \begin{document}  
>>  
>> \section*{Exercise 4.7 - Other Types of Float}  
>>  
>> \subsection*{Standard figure environment}  
>>  
>> \begin{figure}[htbp]  
>> \centering  
>> \includegraphics[width=0.4\textwidth]{example-image}  
>> \caption{Standard figure float}  
>> \label{fig:standard}  
>> \end{figure}
```

Рис. 24: Принудительное размещение

Упражнение 4.7 — пользовательские float

Определение новых типов: photo, diagram

```
(c:/texlive/2025/texmf-dist/tex/context/base/mkii/supp-pdf.mkii
[Loading MPS to PDF converter (version 2006.09.02).]
) (c:/texlive/2025/texmf-dist/tex/latex/epstopdf-pkg/epstopdf-base.sty
(c:/texlive/2025/texmf-dist/tex/latex/latexconfig/epstopdf-sys.cfg))
[1{c:/texlive/2025/texmf-var/fonts/map/pdftex/updmap/pdftex.map} <c:/texlive/20
25/texmf-dist/tex/latex/mwe/example-image.pdf> <c:/texlive/2025/texmf-dist/tex/
latex/mwe/example-image-a.pdf> <c:/texlive/2025/texmf-dist/tex/latex/mwe/exampl
e-image-b.pdf>]
[2] (./exercise-4-7.aux)

LaTeX Warning: Label(s) may have changed. Rerun to get cross-references right.

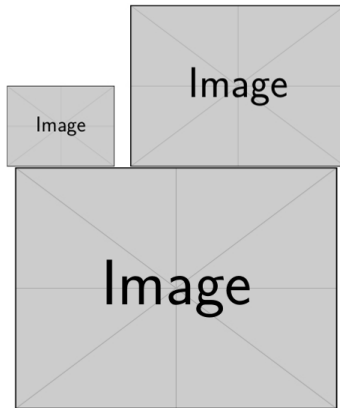
) <c:/Users/ousma/.texlive2025/texmf-var/fonts/pk/ljfour/jknappen/ec/ectt1000.
600pk> <c:/Users/ousma/.texlive2025/texmf-var/fonts/pk/ljfour/jknappen/ec/ecrm1
000.600pk> <c:/Users/ousma/.texlive2025/texmf-var/fonts/pk/ljfour/jknappen/ec/e
cbx1200.600pk> <c:/Users/ousma/.texlive2025/texmf-var/fonts/pk/ljfour/jknappen/
ec/ecbx1440.600pk><c:/texlive/2025/texmf-dist/fonts/type1/public/lm/lmss17.pfb>
Output written on exercise-4-7.pdf (2 pages, 47161 bytes).
```

Рис. 25: Создание документа

Exercise 4.1 - Altering Graphic Appearance

Using example-image as in the documentation

Width controls:



Rotation examples:

Exercise 4.7 - Other Types of Float

Standard figure environment



Figure 1: Standard figure float

Custom photo float



Photo 1: Custom photo float (using trivfloat)

Custom diagram float



Diagram 1: Custom diagram float

Creating custom floats:

```
\usepackage{trivfloat}
```

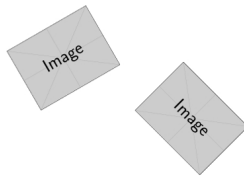
1

```
\trivfloat{photo}  
\trivfloat{diagram}
```

Модифицированное упражнение

4.1

Модифицированное упражнение 4.1

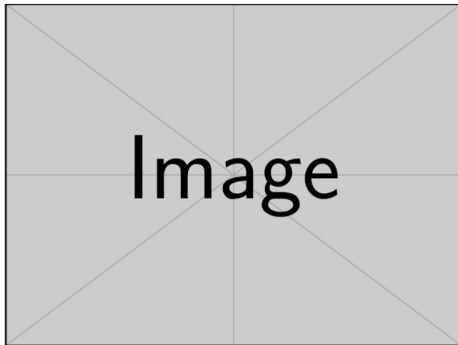


Scaling examples:

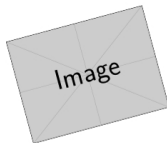


Модифицированное упражнение

4.1



Combined parameters:



Упражнение 4.8 — перекрёстные ссылки

```
PS C:\Users\ousma\Documents\latex-exercises\chapter-4-graphics> @*
>> \documentclass{article}
>> \usepackage[T1]{fontenc}
>> \usepackage{graphicx}
>> \usepackage{hyperref}
>>
>> \begin{document}
>>
>> \section*{Exercise 4.8 - Cross-referencing}
>>
>> \subsection*{Creating labeled figures}
>>
>> \begin{figure}[h]
>> \centering
>> \includegraphics[width=0.4\textwidth]{example-image}
>> \caption{First figure with label}
>> \label{fig:first}
>> \end{figure}
>>
```

Рис. 30: Создание документа

Использование hyperref

```
>> pdflatex exercise-4-8.tex
>> start exercise-4-8.pdf
This is pdfTeX, Version 3.141592653-2.6-1.40.28 (TeX Live 2025) (preloaded format=pdflatex)
 restricted \write18 enabled.
entering extended mode
(./exercise-4-8.tex
LaTeX2e <2025-06-01> patch level 1
L3 programming layer <2025-09-02>
(c:/texlive/2025/texmf-dist/tex/latex/base/article.cls
Document Class: article 2025/01/22 v1.4n Standard LaTeX document class
(c:/texlive/2025/texmf-dist/tex/latex/base/size10.clo))
(c:/texlive/2025/texmf-dist/tex/latex/base/fontenc.sty)
(c:/texlive/2025/texmf-dist/tex/latex/graphics/graphicx.sty)
(c:/texlive/2025/texmf-dist/tex/latex/graphics/keyval.sty)
(c:/texlive/2025/texmf-dist/tex/latex/graphics/graphics.sty)
(c:/texlive/2025/texmf-dist/tex/latex/graphics/trig.sty)
(c:/texlive/2025/texmf-dist/tex/latex/graphics-cfg/graphics.cfg)
(c:/texlive/2025/texmf-dist/tex/latex/graphics-def/pdftex.def)))
(c:/texlive/2025/texmf-dist/tex/latex/hyperref/hyperref.sty
(c:/texlive/2025/texmf-dist/tex/generic/iftex/iftex.sty)
(c:/texlive/2025/texmf-dist/tex/latex/kvsetkeys/kvsetkeys.sty)
(c:/texlive/2025/texmf-dist/tex/generic/kvdefinekeys/kvdefinekeys.sty)
(c:/texlive/2025/texmf-dist/tex/generic/pdfescape/pdfescape.sty
(c:/texlive/2025/texmf-dist/tex/generic/ltxcmds/ltxcmds.sty)
(c:/texlive/2025/texmf-dist/tex/generic/pdftexcmds/pdftexcmds.sty
(c:/texlive/2025/texmf-dist/tex/generic/infwarrerr/infwarrerr.sty)))
(c:/texlive/2025/texmf-dist/tex/latex/hycolor/hycolor.sty)
(c:/texlive/2025/texmf-dist/tex/latex/hyperref/nameref.sty
```

Рис. 31: Компиляция

Exercise 4.8 - Cross-referencing

Creating labeled figures



Figure 1: First figure with label



Figure 2: Second figure with label

Referencing figures in text

As shown in Figure ??, we can see the first example image. Figure ?? shows another example.

We can also reference both figures: see Figures ?? and ??.

Hyperref for clickable links

With the hyperref package, references become clickable links that jump to the corresponding figure.

Label placement rules:

- \label must come after \caption
- Labels should be placed inside the float environment

1

- Use descriptive names: fig:result, tab:data
- Multiple compilations may be needed for references to resolve

Упражнение 4.9 — комплексная работа

```
[1{c:/texlive/2025/texmf-var/fonts/map/pdftex/updmap/pdftex.map} <c:/texlive/2025/texmf-dist/tex/latex/mwe/example-image.pdf> <c:/texlive/2025/texmf-dist/tex/latex/mwe/example-image-a.pdf>]
[2] (./exercise-4-8.aux)

LaTeX Warning: There were undefined references.

LaTeX Warning: Label(s) may have changed. Rerun to get cross-references right.

Package rerunfilecheck Warning: File `exercise-4-8.out' has changed.
(rerunfilecheck)      Rerun to get outlines right
(rerunfilecheck)      or use package `bookmark'.

) <c:/Users/ousma/.texlive2025/texmf-var/fonts/pk/ljfour/jknappen/ec/ectt1000.600pk> <c:/Users/ousma/.texlive2025/texmf-var/fonts/pk/ljfour/jknappen/ec/tcrm1000.600pk> <c:/Users/ousma/.texlive2025/texmf-var/fonts/pk/ljfour/jknappen/ec/ecbx1000.600pk> <c:/Users/ousma/.texlive2025/texmf-var/fonts/pk/ljfour/jknappen/ec/ecrm1000.600pk> <c:/Users/ousma/.texlive2025/texmf-var/fonts/pk/ljfour/jknappen/ec/ecbx1200.600pk> <c:/Users/ousma/.texlive2025/texmf-var/fonts/pk/ljfour/jknappen/ec/ecbx1440.600pk><c:/texlive/2025/texmf-dist/fonts/type1/public/lm/lms17.pfb>
Output written on exercise-4-8.pdf (2 pages, 49190 bytes).
Transcript written on exercise-4-8.log.
```

Рис. 33: Создание

```
PS C:\Users\ousma\Documents\latex-exercises\chapter-4-graphics> @'  
>> \documentclass{article}  
>> \usepackage[T1]{fontenc}  
>> \usepackage{graphicx}  
>> \usepackage{float}  
>> \usepackage{hyperref}  
>>  
>> \graphicspath{{images/}}  
>>  
>> \begin{document}  
>>  
>> \section*{Exercise 4.9 - Comprehensive Graphics Exercise}  
>>  
>> \subsection*{Multiple image types and placements}  
>>  
>> \begin{figure}[t]  
>> \centering  
>> \includegraphics[width=0.45\textwidth]{example-image}  
>> \caption{Top-placed figure}  
>> \label{fig:top}  
>> \end{figure}
```

Рис. 34: Компиляция

```
This is pdfTeX, Version 3.141592653-2.6-1.40.28 (TeX Live 2025) (preloaded format=pdflatex)
 restricted \write18 enabled.
entering extended mode
(./exercise-4-9.tex
LaTeX2e <2025-06-01> patch level 1
L3 programming layer <2025-09-02>
(c:/texlive/2025/texmf-dist/tex/latex/base/article.cls
Document Class: article 2025/01/22 v1.4n Standard LaTeX document class
(c:/texlive/2025/texmf-dist/tex/latex/base/size10.clo))
(c:/texlive/2025/texmf-dist/tex/latex/base/fontenc.sty)
(c:/texlive/2025/texmf-dist/tex/latex/graphics/graphicx.sty)
(c:/texlive/2025/texmf-dist/tex/latex/graphics/keyval.sty)
(c:/texlive/2025/texmf-dist/tex/latex/graphics/graphics.sty)
(c:/texlive/2025/texmf-dist/tex/latex/graphics/trig.sty)
(c:/texlive/2025/texmf-dist/tex/latex/graphics-cfg/graphics.cfg)
(c:/texlive/2025/texmf-dist/tex/latex/graphics-def/pdftex.def)))
(c:/texlive/2025/texmf-dist/tex/latex/float/float.sty)
(c:/texlive/2025/texmf-dist/tex/latex/hyperref/hyperref.sty)
(c:/texlive/2025/texmf-dist/tex/generic/iftex/iftex.sty)
(c:/texlive/2025/texmf-dist/tex/latex/kvsetkeys/kvsetkeys.sty)
(c:/texlive/2025/texmf-dist/tex/generic/kvdefinekeys/kvdefinekeys.sty)
(c:/texlive/2025/texmf-dist/tex/generic/pdfescape/pdfescape.sty)
(c:/texlive/2025/texmf-dist/tex/generic/ltxcmds/ltxcmds.sty)
(c:/texlive/2025/texmf-dist/tex/generic/pdftexcmds/pdftexcmds.sty)
(c:/texlive/2025/texmf-dist/tex/generic/infwarerr/infwarerr.sty)))
(c:/texlive/2025/texmf-dist/tex/latex/hycolor/hycolor.sty)
(c:/texlive/2025/texmf-dist/tex/latex/hyperref/nameref.sty
```

```
[1{c:/texlive/2025/texmf-var/fonts/map/pdftex/updmap/pdftex.map} <c:/texlive/2025/texmf-dist/tex/latex/mwe/example-image.pdf> <c:/texlive/2025/texmf-dist/tex/latex/mwe/example-image-a.pdf>]
[2 <c:/texlive/2025/texmf-dist/tex/latex/mwe/example-image-b.pdf> <c:/texlive/2025/texmf-dist/tex/latex/mwe/example-image-c.pdf>]
[3] (./exercise-4-9.aux)

LaTeX Warning: There were undefined references.

LaTeX Warning: Label(s) may have changed. Rerun to get cross-references right.

Package rerunfilecheck Warning: File `exercise-4-9.out' has changed.
(rerunfilecheck)          Rerun to get outlines right
(rerunfilecheck)          or use package `bookmark'.

) <c:/Users/ousma/.texlive2025/texmf-var/fonts/pk/ljfour/jknappen/ec/tcrm1000.600pk> <c:/Users/ousma/.texlive2025/texmf-var/fonts/pk/ljfour/jknappen/ec/ecbx1000.600pk> <c:/Users/ousma/.texlive2025/texmf-var/fonts/pk/ljfour/jknappen/ec/ecbx1200.600pk> <c:/Users/ousma/.texlive2025/texmf-var/fonts/pk/ljfour/jknappen/ec/ecbx1440.600pk> <c:/Users/ousma/.texlive2025/texmf-var/fonts/pk/ljfour/jknappen/ec/ecrm1000.600pk><c:/texlive/2025/texmf-dist/fonts/type1/public/lm/lmss17.pfb>
Output written on exercise-4-9.pdf (3 pages, 50436 bytes).
Transcript written on exercise-4-9.log.
PS C:\Users\ousma\Documents\latex-exercises\chapter-4-graphics> _
```


Итоги работы

В ходе лабораторной были освоены:

- вставка изображений различных типов
- масштабирование, поворот, комбинирование параметров
- управление размещением плавающих объектов
- создание собственных float-окружений
- работа с перекрёстными ссылками и гиперссылками
- организация структуры каталогов для хранения графики